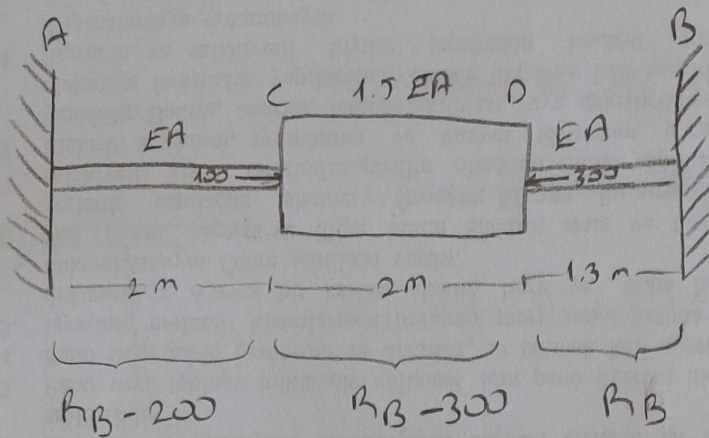


HOMEWORK II



$$R_B - 200 - R_A = 0$$

$$\Delta L_{AC} + \Delta L_{CD} + \Delta L_{DB} = 0$$

$$\frac{(R_B - 200) \cdot 2}{2000} + \frac{(R_B - 300) \cdot 2}{3000} + \frac{R_B \cdot 1.3}{2000} = 0$$

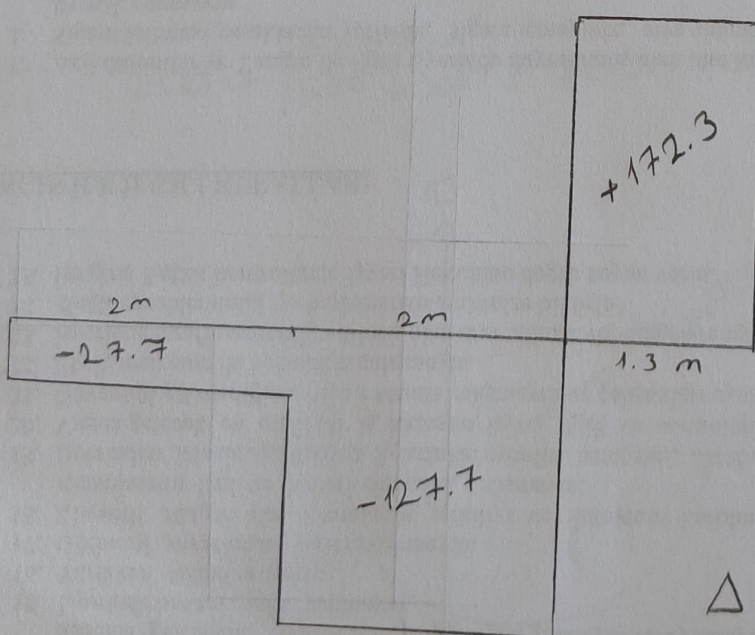
$$2.31 R_B - 398 = 0$$

$$R_B = 172.3$$

$$R_B - 200 = R_A$$

$$R_A = -27.7$$

→ yönü < 0

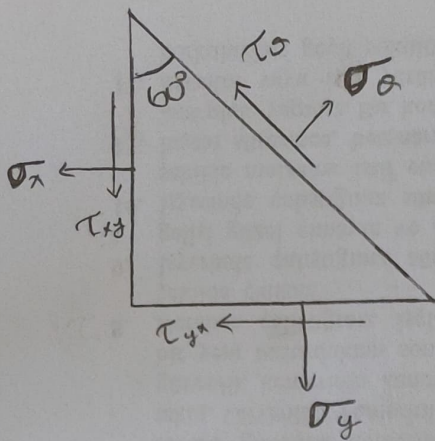


$$\Delta L_{AC} = \frac{(172.3 - 200) \cdot 2}{2000}$$

Sene ERSOY
060200434
sena

$$\Delta L_{AC} = -0.02 \text{ m}$$

$$P_{AC} = 1.98 \text{ m}$$



$$\sigma_{\theta} = \sigma_x \cdot \cos^2 \theta + \sigma_y \sin^2 \theta + 2 \tau_{xy} \sin \theta \cos \theta$$

$$\tau_{\theta} = -(\sigma_x - \sigma_y) \sin \theta \cos \theta + \tau_{xy} (\cos^2 \theta - \sin^2 \theta)$$

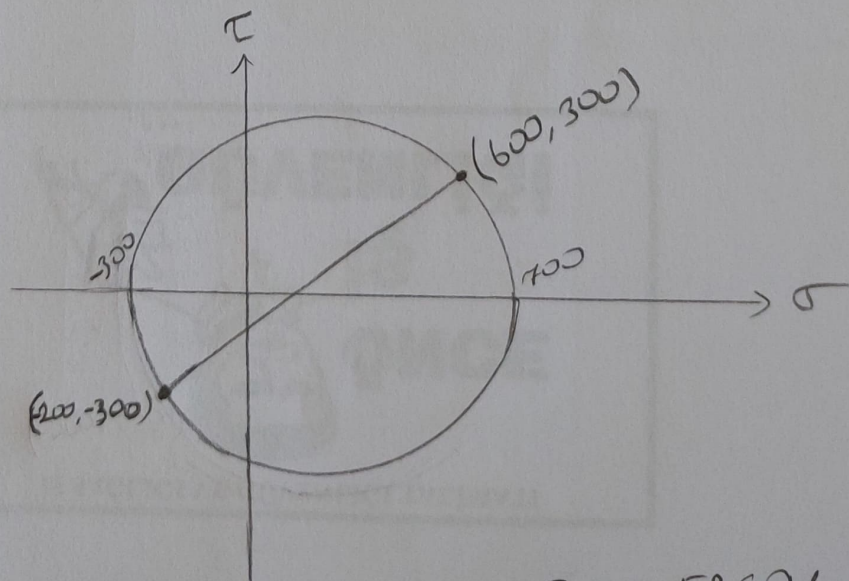
$$\sigma_1 = \frac{1}{2} (\sigma_x + \sigma_y) + \sqrt{\left[\frac{1}{2} (\sigma_x - \sigma_y) \right]^2 + \tau_{xy}^2}$$

$$\sigma_2 = \frac{1}{2} (\sigma_x + \sigma_y) - \sqrt{\left[\frac{1}{2} (\sigma_x - \sigma_y) \right]^2 + \tau_{xy}^2}$$

$$\begin{aligned} \sigma_{60} &= 600 \cdot \frac{1}{4} - 200 \cdot \frac{3}{4} + 2 \cdot 300 \frac{\sqrt{3}}{4} \\ &= 259.8 \end{aligned}$$

$$\begin{aligned} \tau_{60} &= -(600 + 200) \cdot \frac{\sqrt{3}}{4} + 300 \left(-\frac{1}{2} \right) \\ &= -496.4 \end{aligned}$$

$$\sigma_1 = 200 + \sqrt{400^2 + 300^2} = 700 \quad \sigma_2 = 200 - 500 = -300$$



Sena ERSOY
01.02.2023
sena